

# Route decision: Shanghai Global Distribution Center → Rotterdam Distripark

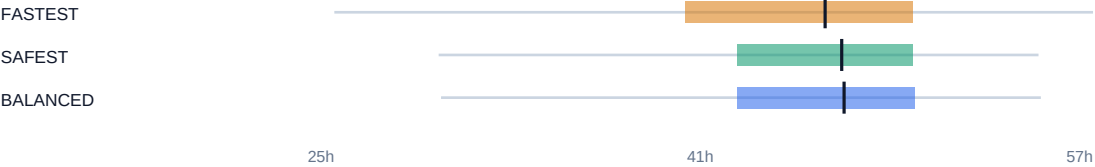
HUB-SHANGHAI → HUB-ROTTERDAM · mode any (STRICT) · cargo perishable\_urgent · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC

**Recommendation: BALANCED**, AIR via direct. Expected arrival in **42 h** (1.7 days); plan for 46 h (p85), worst case 49 h (p95). Cost: freight CN¥898,559. Peak leg risk 2%.

## Options

| Option   | Mode | Via    | Plan h | p50 h | p85 h | p95 h | Freight    | Risk |
|----------|------|--------|--------|-------|-------|-------|------------|------|
| FASTEST  | AIR  | direct | 25     | 40    | 46    | 49    | CN¥932,752 | 2%   |
| SAFEST   | AIR  | direct | 29     | 42    | 46    | 49    | CN¥936,892 | 2%   |
| BALANCED | AIR  | direct | 29     | 42    | 46    | 49    | CN¥898,559 | 2%   |

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



## Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 39.7 h (p85 45.6 h, p95 49.3 h), CN¥932,752 freight, 2 handoffs. 2.2 h faster than the next option. CN¥34,193 more (freight) than BALANCED.

| # | Mode    | To                            | km    | Hours | Delay p85 | Cost       | Risk | Signal |
|---|---------|-------------------------------|-------|-------|-----------|------------|------|--------|
| 1 | ROAD    | Shanghai Pudong International | 30    | 0.4   | +3.6      | CN¥241.78  | 0%   |        |
| 2 | handoff | Shanghai Pudong International |       | 6.0   | +0.0      | CN¥1,008   | 2%   |        |
| 3 | AIR     | Frankfurt Airport             | 8,858 | 11.1  | +8.8      | CN¥893,097 | 0%   |        |
| 4 | AIR     | Amsterdam Airport Cargo       | 366   | 0.5   | +8.8      | CN¥36,880  | 0%   |        |
| 5 | handoff | Amsterdam Airport Cargo       |       | 6.0   | +0.0      | CN¥1,008   | 2%   |        |
| 6 | ROAD    | Rotterdam Distripark          | 64    | 0.9   | +3.6      | CN¥516.42  | 0%   |        |

Transit 12.9 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 24.9 h. Model delay buffer p50/p85/p95: +14.7 / +24.9 / +32.0 h. Route band p50 39.7 h, p85 45.6 h, p95 49.3 h (independent legs); 56.9 h if all legs are late together. Costs: linehaul CN¥930,736, transfers CN¥2,016, disruption surcharge CN¥0.00.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.3 h, p95 49.3 h), CN¥936,892 freight, 2 handoffs. 2.2 h slower than FASTEST. CN¥38,333 more (freight) than BALANCED.

| # | Mode    | To                            | km    | Hours | Delay p85 | Cost       | Risk | Signal |
|---|---------|-------------------------------|-------|-------|-----------|------------|------|--------|
| 1 | ROAD    | Shanghai Pudong International | 30    | 0.4   | +3.6      | CN¥241.78  | 0%   |        |
| 2 | handoff | Shanghai Pudong International |       | 6.0   | +0.0      | CN¥1,008   | 2%   |        |
| 3 | AIR     | London Heathrow Airport       | 9,241 | 11.6  | +8.8      | CN¥931,728 | 0%   |        |
| 4 | handoff | London Heathrow Airport       |       | 6.0   | +0.0      | CN¥1,008   | 2%   |        |
| 5 | ROAD    | Calais Channel Tunnel Hub     | 169   | 2.5   | +3.6      | CN¥1,364   | 0%   |        |
| 6 | ROAD    | Rotterdam Distripark          | 191   | 2.8   | +3.6      | CN¥1,541   | 0%   |        |

Transit 17.3 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.3 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.3 h, p95 49.3 h (independent legs); 54.6 h if all legs are late together. Costs: linehaul CN¥934,875, transfers CN¥2,016, disruption surcharge CN¥0.00.

Why the slowest leg (AIR-SHANGHAI → AIR-HEATHROW, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



BALANCED: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.4 h, p95 49.4 h), CN¥898,559 freight, 2 handoffs. 2.2 h slower than FASTEST. CN¥34,193 cheaper (freight) than the lowest-cost alternative.

| # | Mode    | To                            | km    | Hours | Delay p85 | Cost       | Risk | Signal |
|---|---------|-------------------------------|-------|-------|-----------|------------|------|--------|
| 1 | ROAD    | Shanghai Pudong International | 30    | 0.4   | +3.6      | CN¥241.78  | 0%   |        |
| 2 | handoff | Shanghai Pudong International |       | 6.0   | +0.0      | CN¥1,008   | 2%   |        |
| 3 | AIR     | Frankfurt Airport             | 8,858 | 11.1  | +8.8      | CN¥893,097 | 0%   |        |
| 4 | handoff | Frankfurt Airport             |       | 6.0   | +0.0      | CN¥1,008   | 2%   |        |
| 5 | ROAD    | Duisburg Intermodal Terminal  | 200   | 2.9   | +3.6      | CN¥1,616   | 0%   |        |
| 6 | ROAD    | Rotterdam Distripark          | 197   | 2.9   | +3.6      | CN¥1,588   | 0%   |        |

Transit 17.4 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.4 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.4 h, p95 49.4 h (independent legs); 54.7 h if all legs are late together. Costs: linehaul CN¥896,543, transfers CN¥2,016, disruption surcharge CN¥0.00.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.